

CSCE 679 DATA VISUALIZATION · MODULE 06 · LECTURE NOTES

Multiple Views and Interaction

More than fits in one figure.

MODULE OUTCOMES

- Explain small multiples and when to use them.
- Explain coordinated views and linked selection.
- Explain the interaction vocabulary.
- Explain what interaction costs.
- Design a multi-view figure for a stated task.

1 Small multiples

SMALL MULTIPLES PUT A CATEGORICAL VARIABLE ON SPATIAL REGION, WHICH IS THE MOST ACCURATE CHANNEL FOR IT

Repeat the same chart once per category, arranged in a grid — which puts the categorical variable on spatial region, **rank one for categorical data** (Module 02 §2), **with no capacity limit** where colour had five to seven.

And the comparison works because every panel shares the encoding: the same axes, the same scales, the same mark type, the same colour — so a reader learns the chart once and then reads it *n* times, which is why a grid of twenty panels is not twenty times the work.

Which requires the scales to be shared. Panels with independent axes cannot be compared at all — the whole benefit disappears — and **independent scaling is a common and self-defeating default in plotting libraries**, which scale each facet to its own data.

And the ordering of the panels is a free channel — sort them by something meaningful rather than alphabetically, **and the grid itself carries information** (Module 03 §4's sorting argument, applied to layout rather than to an axis).

- **When you have more than five to seven categories** — past colour's capacity (Module 02 §2) — which is the commonest trigger and the clearest.
- **When the series overlap substantially** and would occlude one another in a single superimposed panel, making the lower ones unreadable.
- **When the reader's task is 'which of these is different', which the grid supports directly** — an odd panel is visible at a glance — **and a superimposed chart does not.**

- **But not when the task is a precise pairwise comparison — position on unaligned scales is rank two rather than rank one** (Module 02 §1), **so superimposing in one panel wins there** and the faceting costs accuracy.
- **Which is the trade: superimpose for precise comparison, facet for everything else** — and it **follows directly from ranks one and two of the quantitative ranking** rather than from any separate principle, which is a satisfying consequence of having the ranking.

2 Coordinated views

SEVERAL VIEWS WITH LINKED SELECTION SHOW WHAT NO SINGLE ENCODING CAN

No single encoding supports every task (Module 01 §2's callout) — so **show several views, each one good at a different task, and link the selection between them.**

Which means that selecting points in a scatter plot highlights the same records in a histogram, on a map, and in a table — and **the reader sees how that selection sits within every other distribution**, which is information no single figure contains.

And brushing is the operation that makes it work: select a region by dragging in one view and see the corresponding marks highlighted everywhere else, which answers 'what are those outliers' directly rather than requiring a separate query.

With one design requirement that is frequently violated: the views must be visible simultaneously — **a linked view behind a tab is not linked**, because the comparison requires both to be in view at once and switching reintroduces the memory demand (CSCE 671 Module 03 §2).

SELECT	mark a subset, persistently
BRUSH	select by dragging a region, and see the selection reflected in the other views
FILTER	remove data from view, which changes the scales unless you deliberately fix them
ZOOM / PAN	change the spatial extent shown
AGGREGATE	change the level of detail shown
REORDER	change the arrangement of a categorical axis (Module 03 section 4)
ENCODE	change the channel assignment itself, which is the most powerful operation and the least often provided

AND NAVIGATE: overview first, zoom and filter, then details on demand -- which is the classical sequence and is still the right default for an unfamiliar dataset.

Overview first, zoom and filter, details on demand — Shneiderman's sequence — **and it is still the right default for exploring something unfamiliar**, because it establishes the context before the detail and so prevents reading a subset as the whole. **The vocabulary makes interaction designable rather than**

improvised: 'add brushing between these two views' is a specification, where 'make it interactive' is not. **And note that filter changes the scales unless you fix them**, which silently breaks comparison across filter states.

3 What interaction costs

- **Nothing is seen until somebody acts. An interaction is a question the reader has to think to ask, and most readers will not ask it — so the default view carries almost all of the communication, whatever else is available.**
- **It does not survive the medium. A printed, pasted, screenshotted, or embedded-in-a-slide figure is its default state — which is how most figures are actually seen, regardless of where they were published.**
- **And discoverability is a real problem — an unsignified interaction is not there at all for anybody who does not try it (CSCE 671 Module 01 §3's signifier argument).**
- **Plus the implementation and maintenance cost, which is substantial compared with a static figure — and which includes keeping the interaction working as the data and the platform change.**
- **So: make the default view answer the main question, and let the interaction answer the follow-up questions. Make the default view answer the main question — because the default is what most readers will ever see, which is CSCE 671 Module 05 §3's default-dominance argument arriving in this subject.**

AND ANIMATION IS A WEAK CHANNEL FOR COMPARISON

Animating between two states requires the reader to hold the first state in memory while seeing the second — which is CSCE 671 Module 03 §2's four-chunk limit, and means an animated transition cannot support a precise comparison of more than a few marks.

So small multiples beat animation for comparing states, reliably and in measured studies — because the states are simultaneously visible and the comparison becomes a spatial judgement rather than a memory one.

But animation is genuinely good for showing a *transition* — object constancy, so that the reader can follow which particular mark moved where — which no static figure conveys and which matters when the identity of the moving items is the point.

Which gives a clean rule: animate to show how something changed, and facet to compare what it was — two different tasks, and the common mistake is animating for comparison, which looks impressive and communicates less than a grid.

4 Designing it

- **List the reader's tasks, and assign each one to a view whose encoding suits it — which is Module 01 §2 applied several times over rather than a new principle.**
- **Share the scales wherever comparison across views is intended, and say so explicitly where they are not shared — since a reader will assume they are and compare anyway.**
- **Link the selection, because an unlinked set of views is a dashboard rather than a tool (see the closing callout) and loses the main benefit of having several views at all.**

- **Keep the number of views small — four is already a lot, and each one costs attention** that is then unavailable for the data (CSCE 671 Module 03 §1).
- **And test what readers actually extract** (Module 13), because **a multi-view figure is very much easier to misread than a single chart** — readers misattribute selections, miss the linking, and compare across unshared scales. **An unlinked set of views is a dashboard rather than a tool, and a dashboard answers whatever questions its designer anticipated**, which is the limitation worth knowing before building one.

AND THE HONEST POSITION ON DASHBOARDS

A dashboard is a set of views chosen in advance for questions anticipated in advance — which is genuinely useful for monitoring and poor for investigation, and those are different activities that get conflated.

So it answers 'is anything unusual' well and 'why is this unusual' badly, because the follow-up question was not anticipated and the dashboard cannot be re-encoded to address it.

Which is exactly why linked selection and the ability to re-encode matter so much — they are what turn a dashboard into something you can actually investigate with, and the ENCODE operation in §2's vocabulary is the most powerful and least often provided for this reason.

And it is CSCE 701 Module 12's measurement problem in visual form: the dashboard makes the measured portion look like the whole, and what was not charted is invisible rather than absent — which is why a dashboard needs a statement of what it does not show, exactly as a metrics programme does.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Munzner, chapters 12 through 14 (slides free)	§1 and §2 — faceting, superimposing, and the view coordination design space.
Shneiderman — The Eyes Have It (free)	§2's task taxonomy and the overview-first sequence , in the original.
Robertson et al. — Effectiveness of Animation in Trend Visualization (free)	§3's callout, measured — small multiples against animation for comparison, with the numbers.
Heer & Shneiderman — Interactive Dynamics for Visual Analysis (free)	§2's vocabulary , organised as a taxonomy of interaction operations.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Facet a chart by a ten-category variable**, and compare against colouring by it.
- 2 **Let the facets scale independently** and show that the comparison breaks.
- 3 **Sort the facets meaningfully** and report what the ordering now conveys.
- 4 **Find a task where superimposing beats faceting**, and justify it from the ranking.
- 5 **Build two linked views with brushing**, and use them to identify an outlier's other properties.
- 6 **Put one view behind a tab** and report what is lost.
- 7 **Show the same comparison as an animation and as small multiples**, and test five readers on accuracy.
- 8 **Make an interactive figure's default state answer the main question**, and screenshot it to check.
- 9 **Count the views in a dashboard you use**, and list the questions it cannot answer.
- 10 **Add a re-encode control** to one figure, and see whether anybody uses it.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why do small multiples work so well?
- 2 What do they require, and what is the common library default that breaks it?
- 3 When do small multiples beat a dense chart, and when not?
- 4 State the superimpose-or-facet rule and its basis.
- 5 What do coordinated views provide, and what is brushing?
- 6 Why is a linked view behind a tab not linked?
- 7 Name seven interaction operations and the classical sequence.
- 8 Give four costs of interaction, and what follows for the default view.
- 9 Why is animation weak for comparison and good for transition?
- 10 What is a dashboard good and bad at, and what makes it investigable?